

# Colour Palettes

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## Introduction

Colour in a data visualisation is not decoration — it encodes information. The right palette makes ordered values read as ordered, signed values read as signed, and categorical values read as distinct without implying a ranking. The wrong palette (the infamous rainbow) creates false visual hotspots where the hue cycle bends and obscures structure that a perceptually uniform palette would reveal. Mastering the three palette families and choosing among them deliberately is one of the highest-payoff skills in figure design.

## Prerequisites

A working knowledge of `ggplot2`'s colour and fill scales, and an awareness of colour-vision deficiencies that affect roughly 8 % of male readers.

## Theory

Three palette families serve three different data types:

- **Sequential:** monotone hue or saturation gradient from low to high (`viridis`, `plasma`, `magma`, `YlOrRd`, `Blues`); for ordered data without a meaningful zero or centre.
- **Diverging:** two hues meeting at a neutral midpoint (`RdBu`, `PuOr`, `BrBG`); for signed data where positive and negative deviations from a centre are both meaningful.
- **Qualitative:** distinct, equally salient hues (`Set1`, `Set2`, `Dark2`, `Okabe-Ito`); for unordered categorical groups, ideally limited to eight or fewer.

`viridis` is the modern default for continuous scales because it is perceptually uniform (equal data steps look like equal visual steps), colour-blind safe under deuteranopia and protanopia, and degrades gracefully to greyscale. ColorBrewer palettes label colour-blind-friendly variants explicitly (`display.brewer.all(colorblindFriendly = TRUE)`), and the Okabe-Ito eight-colour palette was designed specifically for accessibility.

## Assumptions

The data type matches the palette family — sequential for ordered values, diverging for signed, qualitative for nominal.

## R Implementation

```
library(ggplot2); library(viridis); library(RColorBrewer)

# Continuous: viridis
ggplot(faithfuld, aes(waiting, eruptions, fill = density)) +
  geom_tile() +
  scale_fill_viridis_c(option = "plasma") +
  theme_minimal()

# Continuous: diverging
```

```

ggplot(mtcars, aes(wt, mpg, colour = hp - mean(hp))) +
  geom_point(size = 3) +
  scale_colour_gradient2(low = "blue", mid = "grey80", high = "red",
                        midpoint = 0) +
  theme_minimal()

# Qualitative: Brewer Set2
ggplot(mtcars, aes(wt, mpg, colour = factor(cyl))) +
  geom_point(size = 3) +
  scale_colour_brewer(palette = "Set2") +
  theme_minimal()

# Show colour-blind-safe palettes
display.brewer.all(colorblindFriendly = TRUE)

```

## Output & Results

Four illustrative figures: a 2D density tile plot using **plasma** for sequential encoding, a centred-at-zero scatter using **gradient2** for signed deviations, a categorical scatter using Brewer **Set2** for cylinder counts, and a panel of all colour-blind-friendly Brewer palettes for reference.

## Interpretation

A reporting sentence (figure caption): “Cell colour encodes 2D kernel density on the **viridis::plasma** perceptually uniform palette; positive and negative horsepower deviations from the mean use a red-blue diverging scale centred at zero.” Always state the palette by name when it is non-default; readers should not have to guess.

## Practical Tips

- Use **viridis** (or its variants **magma**, **plasma**, **inferno**, **cividis**) as the default for continuous scales; all are perceptually uniform and colour-blind safe.
- Limit qualitative palettes to 8 colours or fewer; beyond that, switch to shapes, linetypes, or facets to disambiguate groups.
- Use **scale\_colour\_gradient2()** (or **scale\_fill\_gradient2()**) for any signed quantity; the centred palette communicates symmetry that one-sided sequentials cannot.
- For greyscale fallback (photocopying, monochrome printing), **viridis** translates almost losslessly; rainbow hues collapse into similar greys and become unreadable.
- Verify colour-blind accessibility on every figure with **colorBlindness::cvdPlot()**; simulating deuteranopia and protanopia takes seconds and prevents costly resubmissions.
- For heatmaps, perceptual uniformity is non-negotiable — a non-uniform palette creates apparent hotspots wherever the hue cycle bends.

## R Packages Used

**viridis** for **viridis**, **magma**, **plasma**, **inferno**, **cividis** continuous scales; **RColorBrewer** for the Brewer family of qualitative, sequential, and diverging palettes; **colorspace** for HCL-based custom palettes; **paletteer** for unified access to dozens of curated palettes from across the R ecosystem.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.

- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts